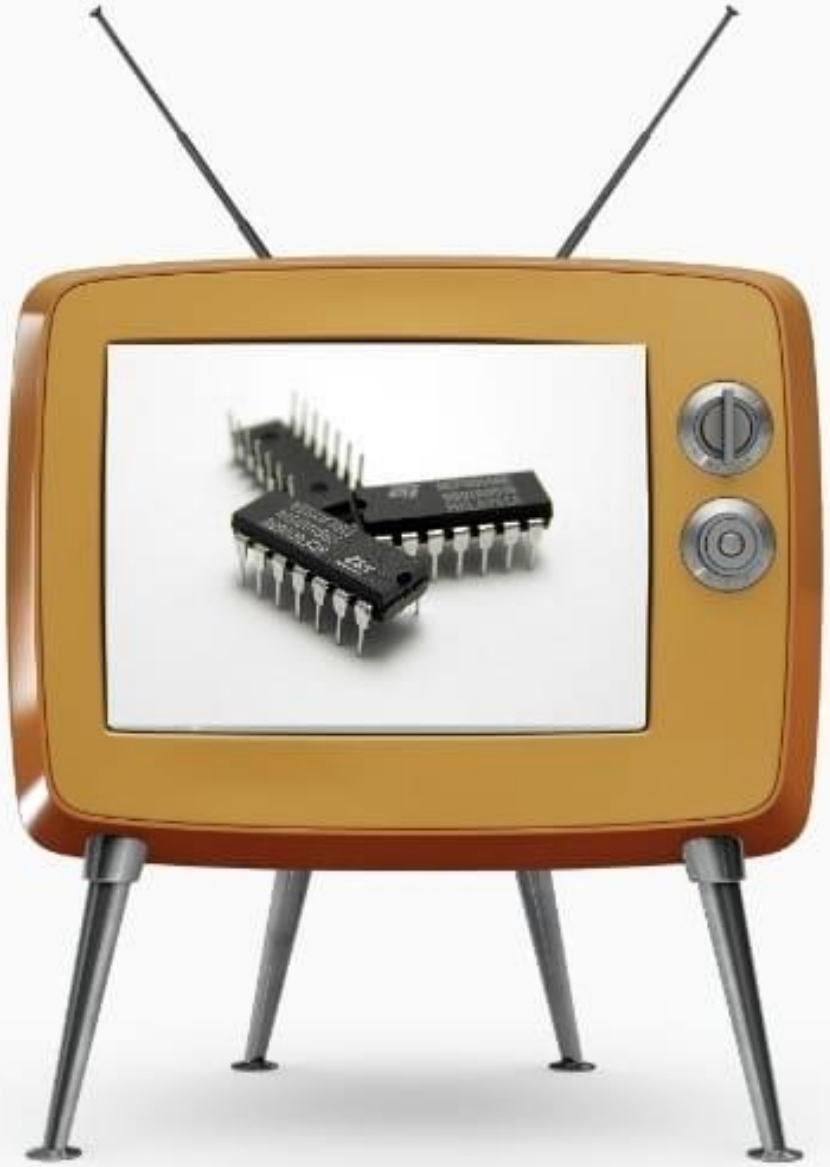
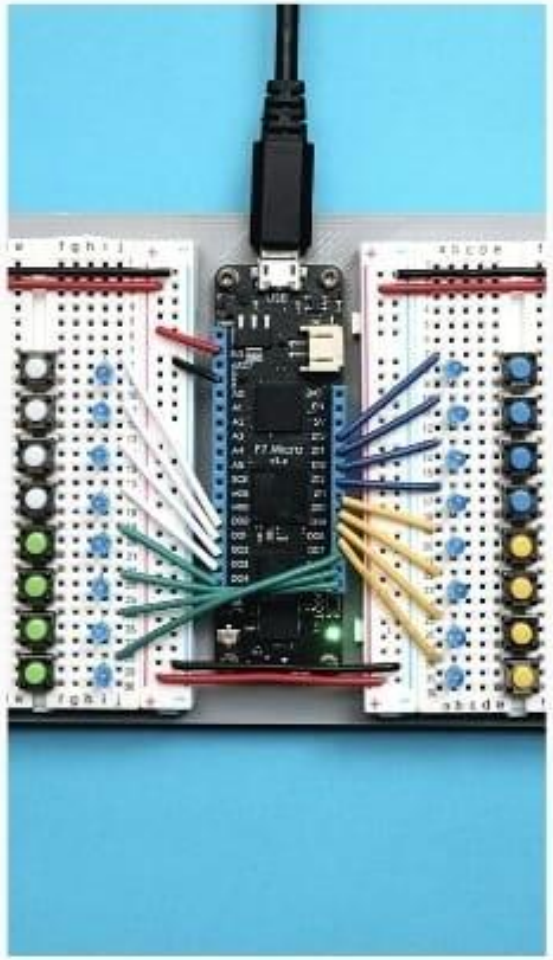
I²C MASTER CONTROLLER USING VERILOG



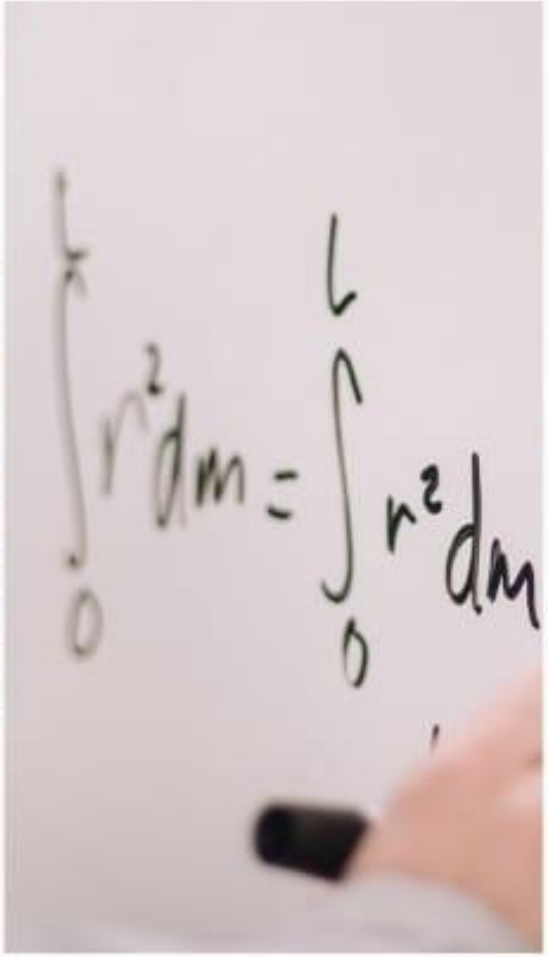
INTRODUCTION TO I2C

- **I^2C (Inter-Integrated Circuit)** is a two-wire serial protocol using **SDA (data)** and **SCL (clock)** lines.
- It enables communication between a **master** and one or more **slave** devices.
- Uses **start/stop conditions**, **7-bit addressing**, and **ACK/NACK** signals for reliable data transfer.
- Supports multiple devices on the same bus, reducing pin usage.
- Speeds range from **100 kbps to 3.4 Mbps** depending on the mode.
- Ideal for **short-distance, low-speed** communication in embedded systems.
- **Applications:** EEPROMs, temperature sensors, real-time clocks (RTC), LCDs, accelerometers, and battery monitors.

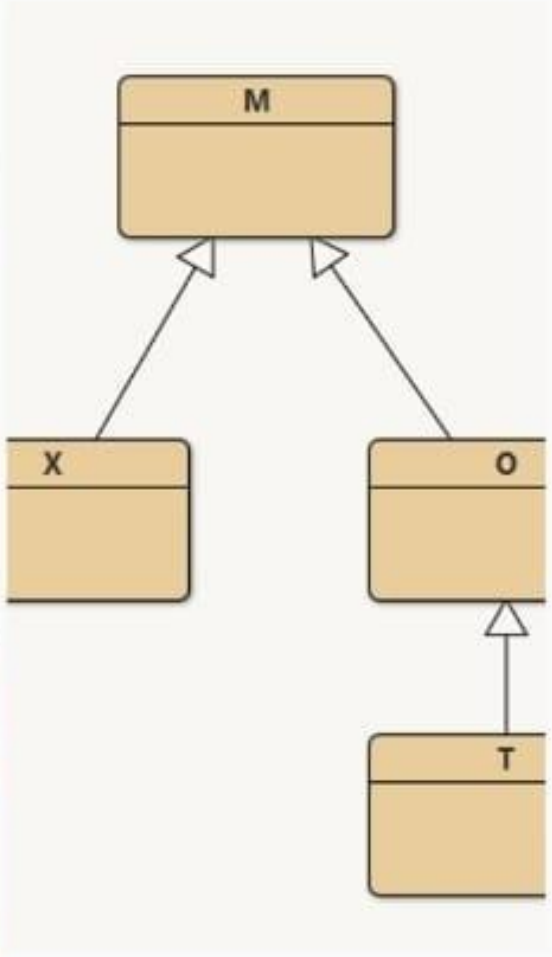
OBJECTIVE



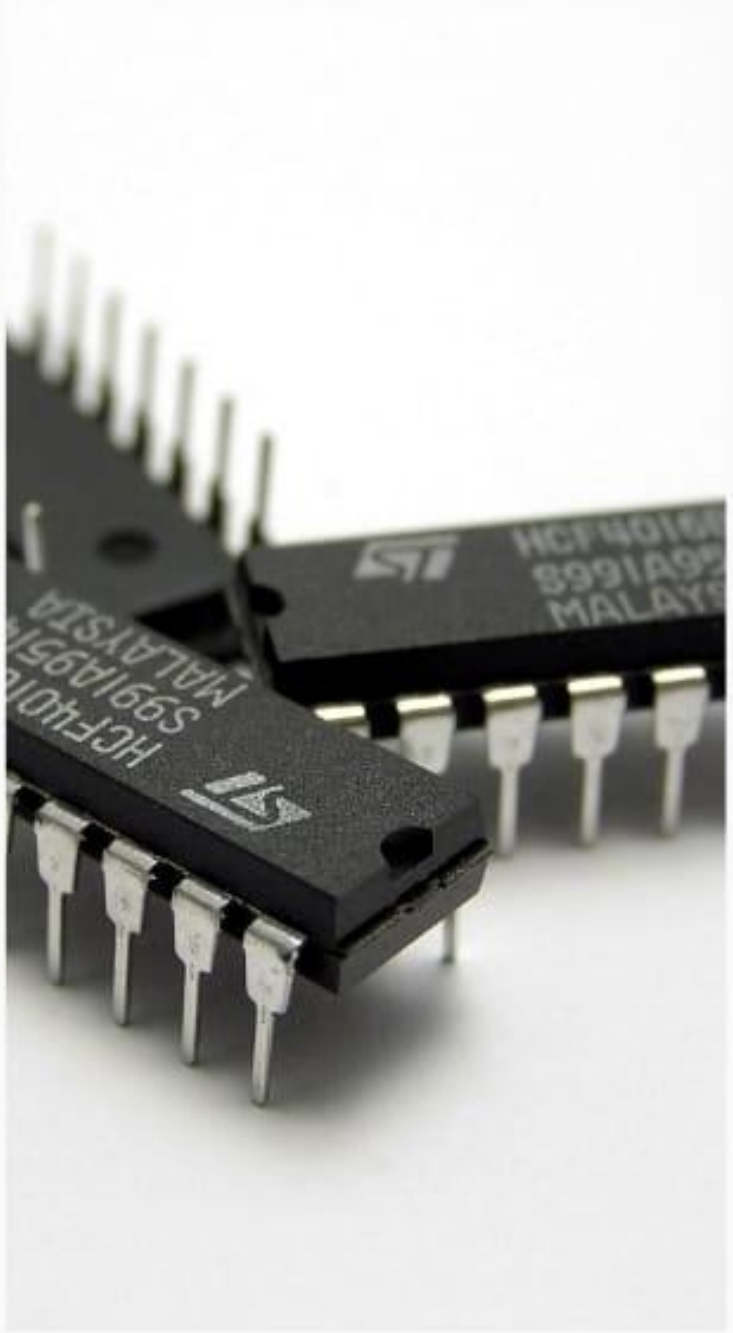
Implement an I²C master controller using Verilog.



Simulate a write operation with address + 1-byte data.



Handle start condition, address transmission, ACK, data write, and stop condition.



INPUTS & FUNCTIONAL CONDITIONS

System clock (`clk`) is essential for operation.

This clock signal synchronizes the data transmission within the I²C protocol.

Reset (`rst`) is active high.

The reset signal initializes the system and ensures all components start from a known state.

Enable signal triggers the I²C operation.

This signal must be activated for the I²C communication process to begin.

7-bit slave address (`addr6:0`) specifies the target device.

This address is used to identify the specific slave device on the I²C bus.

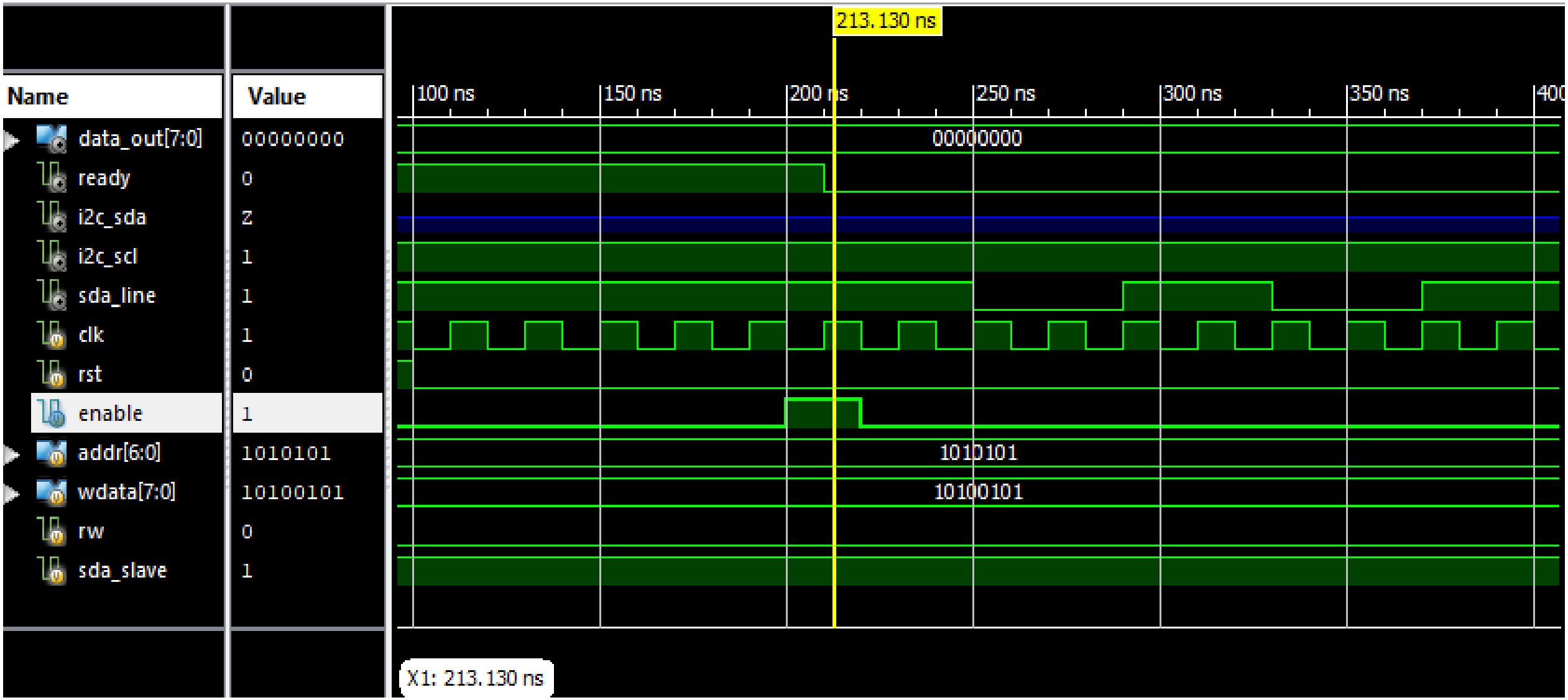
Data to be written is represented by `wdata7:0`.

This is the actual byte of information that will be sent to the slave device.

FSM transitions through defined states: IDLE → START → SEND_ADDR → ACK → WRITE → ACK → STOP.

The Finite State Machine (FSM) controls the flow of the I²C communication process.

OUTPUT WAVEFORM



LIMITATIONS

Timing not compliant with all I²C specs (ideal for learning/demo only).

Single-byte read/write only supported.

No support for repeated start conditions.

No multi-byte transactions allowed.

FUTURE SCOPE

THANK YOU..